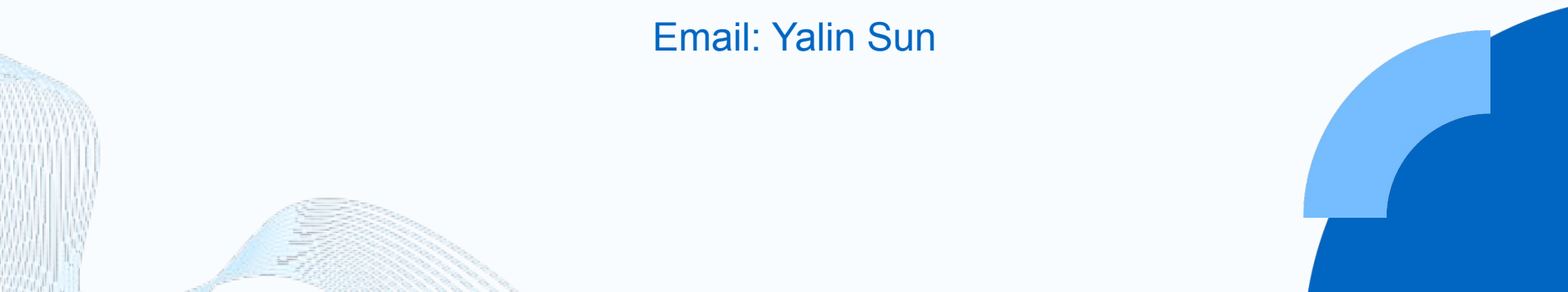




Midterm Mastery Presentation

Present By Yalin Sun

Email: Yalin Sun

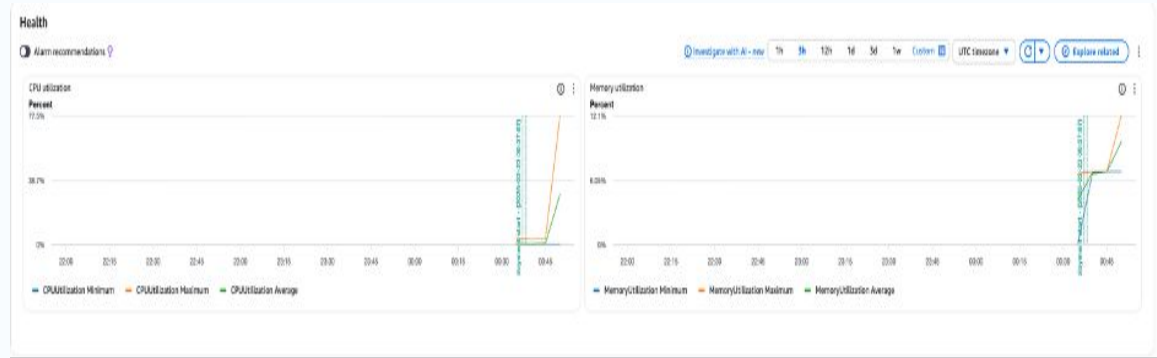


Step I: HW 6 Results

Part II: Identifying Performance Bottlenecks!

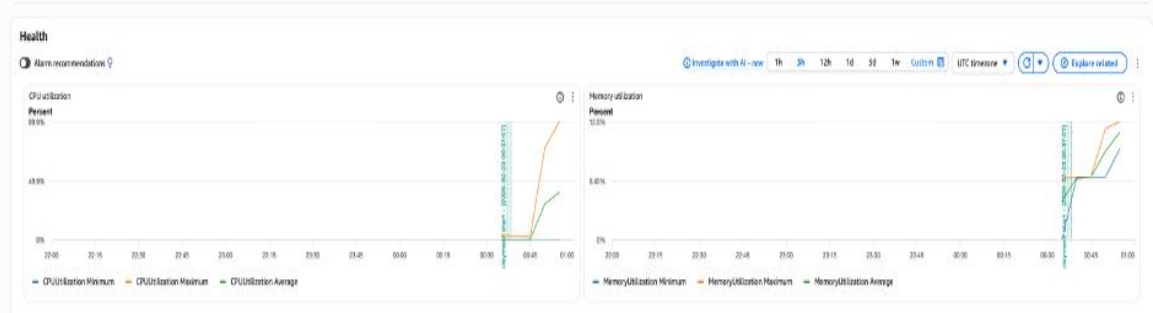
Test 1: Baseline Load

- Users: 5
- Spawn Rate: 5 users/second
- Run Time: 2 minutes



Test 2: Breaking Point

- Users: 20
- Spawn Rate: 20 users/second
- Run Time: 3 minutes

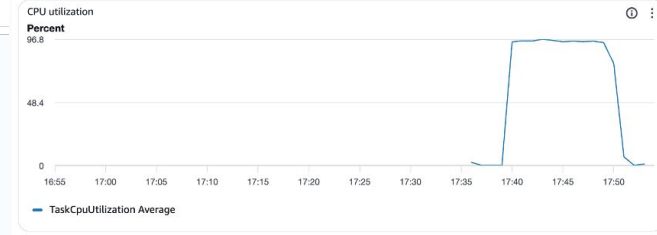
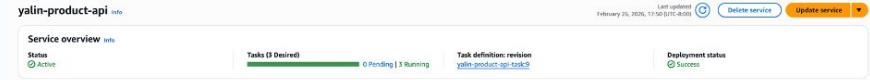
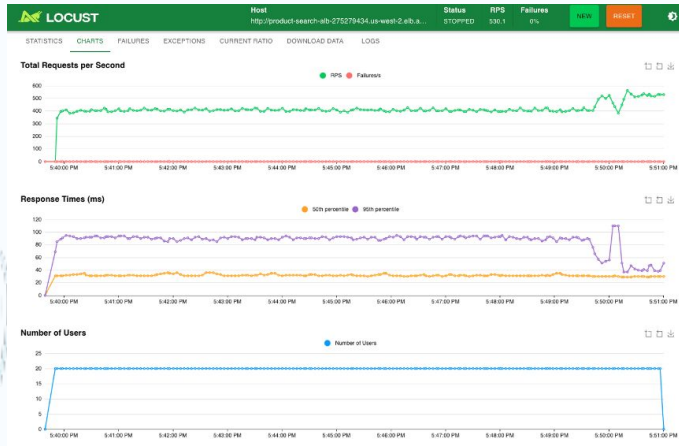


Step I: HW 6 Results

Part II: Horizontal Scaling with Auto Scaling

Test 1: Breaking point

- Users: 20
- Spawn Rate: 20 users/second
- Run Time: 12 minutes

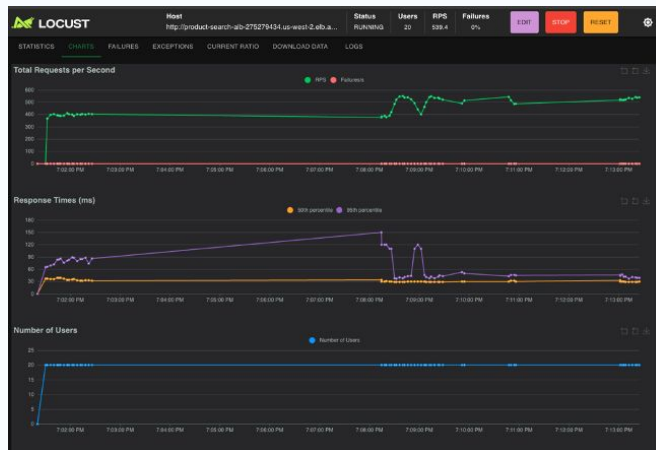
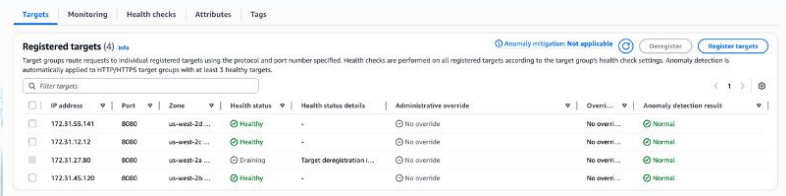
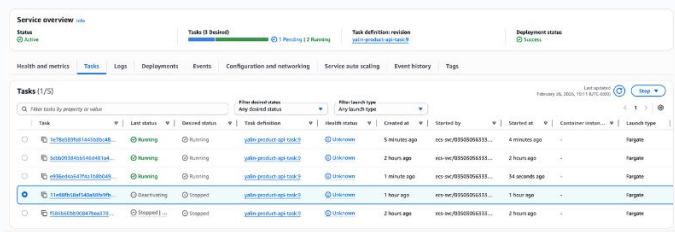


Step I: HW 6 Results

Part II: Horizontal Scaling with Auto Scaling

Test 2: Resilience Test

- Manually Stop a Task





Step II: What did you find valuable so far?

1. Concurrency and Distributed Systems

- Definition / HW3/ MapReduce

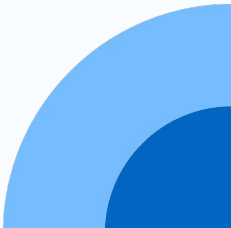
2. Infrastructure as Code and Reproducibility

- Terraform / Docker / ECS, ECR

3. Time, Ordering, and the Illusion of a Global Clock

- Lamport's logical clocks / Race Condition

4. APIs as Contracts Between Systems

- HTTP return codes / REST-style APIs /RPC-style APIs
- 

Step III: Crashing and recovering!

1. Introduction

- “Bad Version” / Good Version

2. How I Design the Failure

- Two load test to show the problem

3. Fix: How I Improved the System

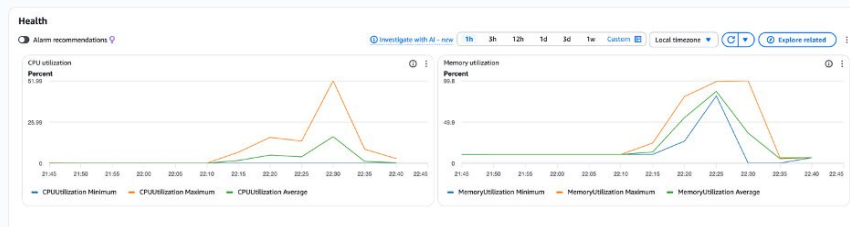
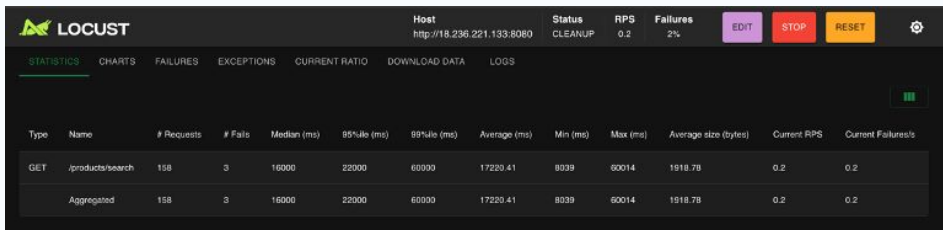
- Fail fast / Limit damage

4. Good Version: Results after Fix

- Latency / CPU usage / CloudWatch Logs

Step III: Crashing and recovering!

1. Bad Version



2. Good Version

